

ASSIGNMENT 14.3

2303A51330

Kashireddy Akshitha
Batch 20

Task 1: AI-Assisted Personal Portfolio Website

Scenario

A student wants to showcase academic projects, technical skills, and contact information through a personal portfolio website. AI tools are used to speed up development while maintaining quality and customization.

Tasks

1. Use GitHub Copilot to generate a semantic HTML structure with sections:
 - o Home
 - o About
 - o Projects
 - o Contact
2. Ask Copilot to suggest responsive CSS using Flexbox or Grid.
3. Customize AI-generated CSS to add:

- o Hover effects on project cards
- o Smooth scrolling navigation

Expected Outcome

- Clean and well-structured portfolio website
- Responsive layout for desktop and mobile screens
- Interactive project cards

PROMPT:

```
<!--Generate a semantic HTML structure with the following sections: Home- About Projects Contact  
Then, suggest responsive CSS using Flexbox or Grid for layout.customize the CSS to include:Hover  
effects on project cards (e.g., scale up slightly, add shadow) Smooth scrolling navigation between  
sections--!>
```

```
<!--Ensure the HTML uses semantic tags (header, nav, section, footer) and the CSS is clean, modern,  
and responsive.--!>
```

CODE:

```
<!DOCTYPE html>  
<html lang="en">  
<head>  
<meta charset="UTF-8">  
<meta name="viewport" content="width=device-width, initial-scale=1.0">  
<title>Akshitha | Portfolio</title>
```

```
<link href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600&display=swap"  
rel="stylesheet">
```

```
<style>
```

```
:root{
```

```
--bg:#0f172a;
--bg2:#1e293b;
--text:#ffffff;
--card:rgba(255,255,255,0.05);
}
```

```
.light{
--bg:#f8fadc;
--bg2:#e2e8f0;
--text:#0f172a;
--card:#ffffff;
}
```

```
*{
margin:0;
padding:0;
box-sizing:border-box;
font-family:'Poppins',sans-serif;
}
```

```
body{
background:linear-gradient(135deg,var(--bg),var(--bg2));
color:var(--text);
scroll-behavior:smooth;
transition:0.4s;
}
```

```
header{
display:flex;
justify-content:space-between;
align-items:center;
padding:20px 60px;
position:sticky;
top:0;
background:rgba(255,255,255,0.05);
backdrop-filter:blur(10px);
z-index:1000;
}
```

```
nav a{
margin:0 15px;
text-decoration:none;
color:var(--text);
transition:0.3s;
}
```

```
nav a:hover{
color:#38bdf8;
}
```

```
.toggle-btn{
cursor:pointer;
font-size:18px;
background:none;
border:none;
color:var(--text);
}
```

```
section{
padding:80px 60px;
opacity:0;
transform:translateY(40px);
transition:all 0.8s ease;
}
```

```
section.show{
opacity:1;
transform:translateY(0);
}
```

```
.hero{
text-align:center;
padding-top:120px;
}
```

```
.hero h2{
font-size:48px;
}
```

```
.role{
color:#38bdf8;
margin-bottom:15px;
}
```

```
.card{
background:var(--card);
padding:25px;
border-radius:15px;
margin-top:20px;
}
```

```
.skills-clean .skill-item{
margin:8px 0;
}
```

```
.skills-clean .divider{
height:1px;
background:rgba(255,255,255,0.1);
margin:10px 0;
}
```

```
.projects{
display:flex;
flex-direction:column;
gap:25px;
}
```

```
.project-card{
padding:20px;
border-radius:15px;
background:var(--card);
transition:0.4s;
}
```

```
.project-card:hover{
transform:translateY(-10px);
}
```

```
.contact-link{
color:var(--text);
text-decoration:none;
}
```

```
.contact-link:hover{
text-decoration:underline;
}
```

```
footer{
text-align:center;
padding:20px;
color:gray;
}
```

```
</style>
</head>
```

```
<body>
```

```
<header>
<h1>Kashireddy Akshitha</h1>
```

```
<nav>
<a href="#about">About</a>
<a href="#skills">Skills</a>
<a href="#projects">Projects</a>
<a href="#education">Education</a>
<a href="#contact">Contact</a>
</nav>
```

```
<button class="toggle-btn" onclick="toggleTheme()" id="themelcon"> 🌙 </button>
</header>
```

```
<section class="hero show">
<h2>Hi, I'm Kashireddy Akshitha <span id="typing"></span></h2>
<div class="role">Computer Science Student | Building and Learning Every Day</div>
<p>Learning by building and improving through practice.</p>
</section>
```

```
<section id="about">
<h2>About Me</h2>
<div class="card">
<p>B.Tech Computer Science student with a strong foundation in Data Structures, Algorithms, and Machine Learning.</p>
<p>Familiar with Python and deep learning frameworks, with hands-on experience building real-time and NLP projects.</p>
<p>Currently learning software engineering practices and seeking internship opportunities.</p>
</div>
</section>
```

```
<section id="skills">
<h2>Technical Skills</h2>
```

<div class="card skills-clean">

<div class="skill-item">Programming: Python • C • SQL</div>

<div class="divider"></div>

<div class="skill-item">Core Concepts: Data Structures • OOP • DBMS</div>

<div class="divider"></div>

<div class="skill-item">Technologies: Machine Learning • NLP • OpenCV</div>

<div class="divider"></div>

<div class="skill-item">Frameworks: PyTorch • TensorFlow</div>

</div>

</section>

<section id="projects">

<h2>Projects</h2>

<div class="projects">

<div class="project-card">

<h3>AI vs Real Image Classification</h3>

<p>Developed a deep learning model using MobileNetV2 to classify AI-generated and real-world images.</p>

<p>Applied transfer learning and achieved ~87% accuracy.</p>

<p>Tech: PyTorch, CNN, Transfer Learning</p>

</div>

<div class="project-card">

<h3>Hybrid Transformer-Capsule Text Classifier</h3>

<p>Built a hybrid NLP model combining Transformer and Capsule Networks.</p>

<p>Captured contextual and hierarchical features improving classification.</p>

<p>Tech: TensorFlow, NLP</p>

</div>

<div class="project-card">

<h3>Real-Time Activity & Posture Detection</h3>

<p>Developed a real-time system to detect human activities using webcam input.</p>

<p>Used pose estimation and smoothing for stable predictions.</p>

<p>Tech: OpenCV, MediaPipe, TensorFlow</p>

</div>

</div>

</section>

<section id="education">

<h2>Education</h2>

<div class="card">

<p>SR University (2023-2027)

B.Tech CSE | CGPA: 8.58</p>


```
<p><strong>Sri Chaitanya Junior College</strong><br>
MPC | 953</p>
</div>
```

```
</section>
```

```
<section id="contact">
<h2>Contact</h2>
```

```
<div class="card">
```

```
<p>
<a href="mailto:akshithakashireddy@gmail.com" class="contact-link">Email</a> |
<a href="https://www.linkedin.com/in/akshithakashireddy/" class="contact-link"
target="_blank">LinkedIn</a> |
<a href="https://leetcode.com/u/kashireddy/" class="contact-link" target="_blank">LeetCode</a>
</p>
```

```
</div>
```

```
</section>
```

```
<footer>
<p>© 2026 Akshitha | Portfolio</p>
</footer>
```

```
<script>
```

```
// typing effect
const text="Akshitha";
let i=0;

function type(){
if(i<text.length){
document.getElementById("typing").innerHTML+=text.charAt(i);
i++;
setTimeout(type,150);
}
}
```

```
type();
```

```
// scroll animation
const sections=document.querySelectorAll("section");
```

```
window.addEventListener("scroll",()=>{
sections.forEach(sec=>{
const top=sec.getBoundingClientRect().top;
if(top<window.innerHeight-100){
sec.classList.add("show");
}
});
});
```

```
// theme toggle
function toggleTheme(){
```

```
document.body.classList.toggle("light");

const icon=document.getElementById("themelcon");

if(document.body.classList.contains("light")){
icon.textContent="☀️";
localStorage.setItem("theme","light");
}else{
icon.textContent="🌙";
localStorage.setItem("theme","dark");
}

}

// load saved theme
window.onload=function(){

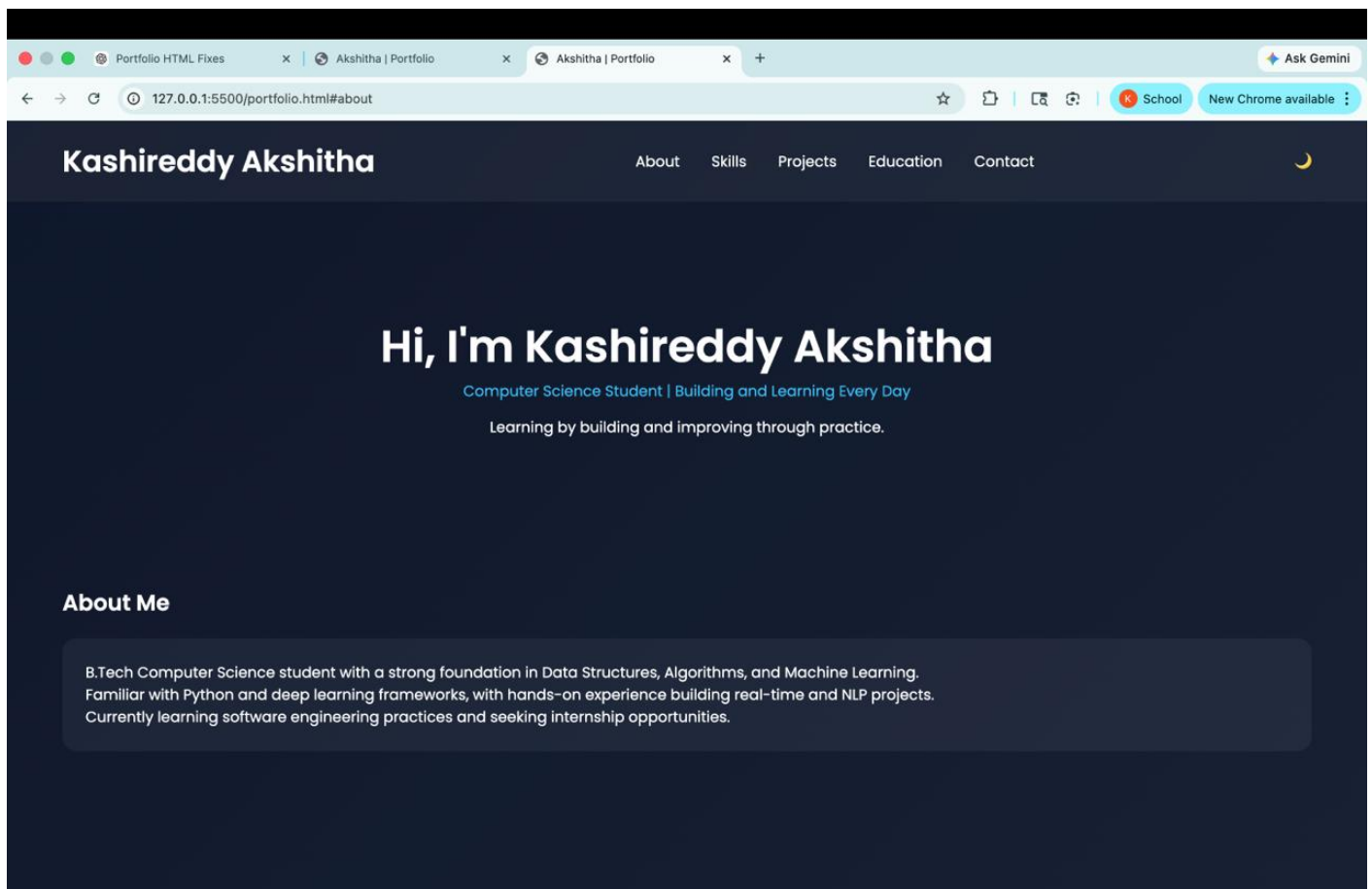
const savedTheme=localStorage.getItem("theme");

if(savedTheme==="light"){
document.body.classList.add("light");
document.getElementById("themelcon").textContent="☀️";
}

};

</script>
</body>
</html>
```

OUTPUT:



Task 2: AI-Generated Restaurant Landing Page

Scenario

A local restaurant requires a simple yet attractive landing page to display its offerings. The developer uses AI assistance to rapidly generate UI components.

Tasks

1. Use Copilot to create a navigation bar with links:

- o Home
- o Menu
- o Gallery

Contact

2. Generate a menu section styled with CSS cards.
3. Implement a JavaScript-based image slider for the gallery using Copilot suggestions.
4. Customize animations or transitions for better user experience.

Expected Outcome

- Fully functional restaurant landing page
- Responsive navigation bar
- Working image slider with smooth transitions

PROMPT:

<!-- create a basic HTML5 structure for a restaurant landing page -->

CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Delicious Bites | Restaurant</title>
  <link href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600&display=swap"
rel="stylesheet">
  <style>
    :root {
      --bg: #fff5e1;
      --text: #333;
      --primary: #ff6347;
      --secondary: #ffcccb;
    }
    * {
      margin: 0;
      padding: 0;
      box-sizing: border-box;
      font-family: 'Poppins', sans-serif;
    }
    body {
      background-color: var(--bg);
      color: var(--text);
    }
    header {
      display: flex;
      justify-content: space-between;
      align-items: center;
      padding: 20px 60px;
      background-color: var(--primary);
```

```

    color: #fff;
}
nav a {
    margin: 0 15px;
    text-decoration: none;
    color: #fff;
    transition: 0.3s;
}
nav a:hover {
    color: var(--secondary);
}
.hero {
    text-align: center;
    padding-top: 120px;
}
.hero h1 {
    font-size: 48px;
}
.hero p {
    margin-top: 10px;
    font-size: 18px;
}
</style>
</head>
<body>

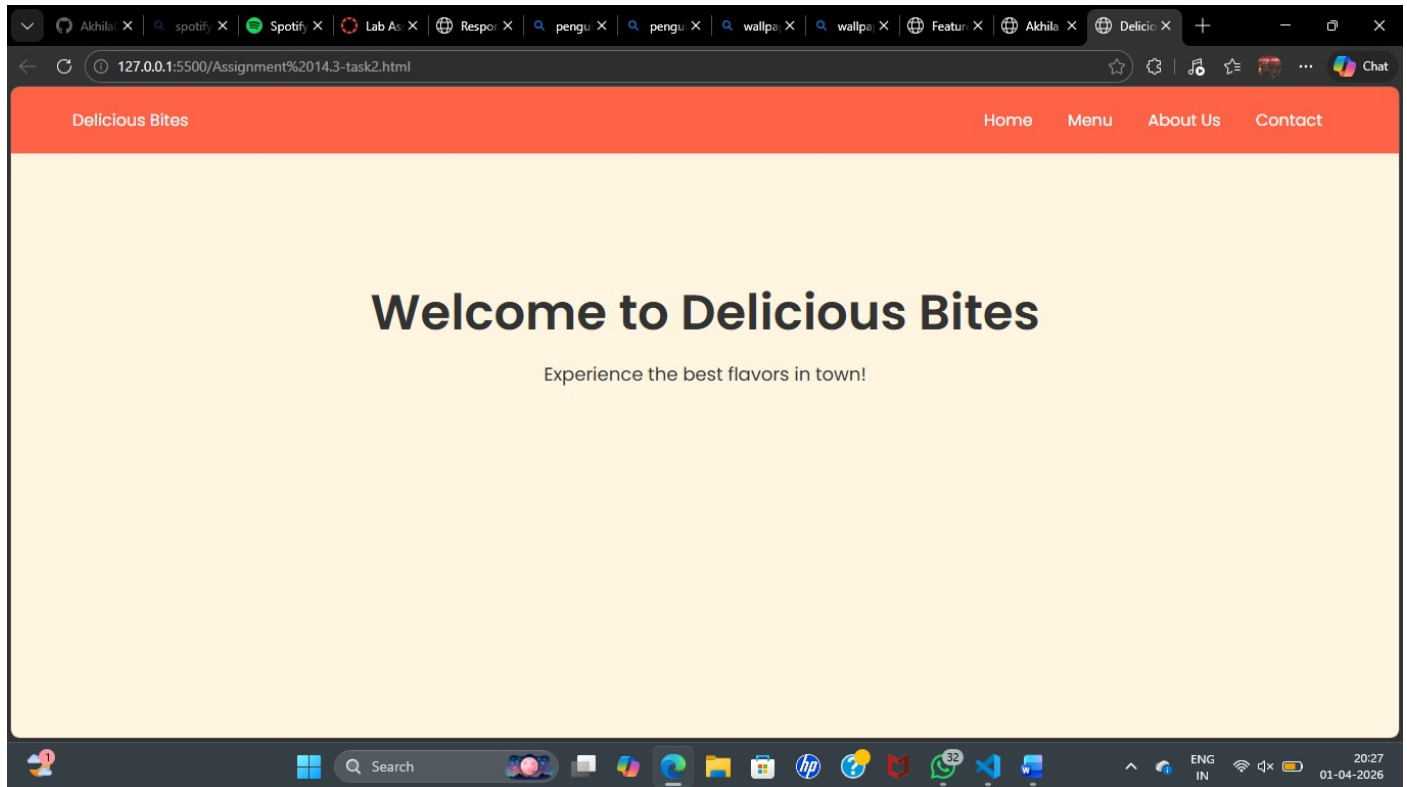
<header>
    <div class="logo">Delicious Bites</div>
    <nav>
        <a href="#home">Home</a>
        <a href="#menu">Menu</a>
        <a href="#about">About Us</a>
        <a href="#contact">Contact</a>
    </nav>
</header>

<section class="hero" id="home">
    <h1>Welcome to Delicious Bites</h1>
    <p>Experience the best flavors in town!</p>
</section>

    <!-- Additional sections for menu, about us, and contact can be added here -->
</body>
</html>

```

OUTPUT:



Task 3: AI-Powered Event Registration Form

Scenario

SR University is hosting a technical fest and requires an online registration form with real-time validation to ensure accurate user input.

Tasks

1. Ask Copilot to generate an HTML form with the following fields:

- o Name
- o Email
- o Phone Number
- o Department
- o Event Selection

2. Use AI assistance to style the form using CSS for readability and accessibility.

3. Implement JavaScript validation using Copilot:

- o Email format validation
- o Phone number length validation
- o Required field checks

Expected Outcome

- User-friendly and visually appealing form
- Real-time validation messages
- Prevention of invalid submissions'

PROMPT:

<!--Create an online registration form for SR University's technical fest with fields for Name, Email,

Phone Number, Department, and Event Selection; style the form using CSS for readability and accessibility; implement JavaScript validation for email format, phone number length, and required field checks; ensure real-time validation messages and prevention of invalid submissions so the form is user-friendly, visually appealing, and fully functional.

-->

CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>SR University Tech Fest Registration</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #f4f4f4;
      display: flex;
      justify-content: center;
      align-items: center;
      height: 100vh;
    }
    .registration-form {
      background-color: #fff;
      padding: 20px;
      border-radius: 5px;
      box-shadow: 0 0 10px rgba(0, 0, 0, 0.1);
      width: 300px;
    }
    .registration-form h2 {
      text-align: center;
      margin-bottom: 20px;
    }
    .registration-form input, .registration-form select {
      width: 100%;
      padding: 10px;
      margin-bottom: 10px;
      border: 1px solid #ccc;
      border-radius: 3px;
    }
    .registration-form button {
      width: 100%;
      padding: 10px;
      background-color: #28a745;
      color: #fff;
```

```

    border: none;
    border-radius: 3px;
    cursor: pointer;
}
.registration-form button:hover {
    background-color: #218838;
}
.error-message {
    color: red;
    font-size: 12px;
}
</style>
</head>
<body>
<form class="registration-form" id="regForm">
    <h2>Tech Fest Registration</h2>
    <input type="text" id="name" placeholder="Name" required>
    <div class="error-message" id="nameError"></div>

    <input type="email" id="email" placeholder="Email" required>
    <div class="error-message" id="emailError"></div>

    <input type="text" id="phone" placeholder="Phone Number" required>
    <div class="error-message" id="phoneError"></div>

    <select id="department" required>
        <option value="">Select Department</option>
        <option value="CSE">Computer Science</option>
        <option value="ECE">Electronics</option>
        <option value="ME">Mechanical</option>
        <!-- Add more departments as needed -->
    </select>
    <div class="error-message" id="deptError"></div>

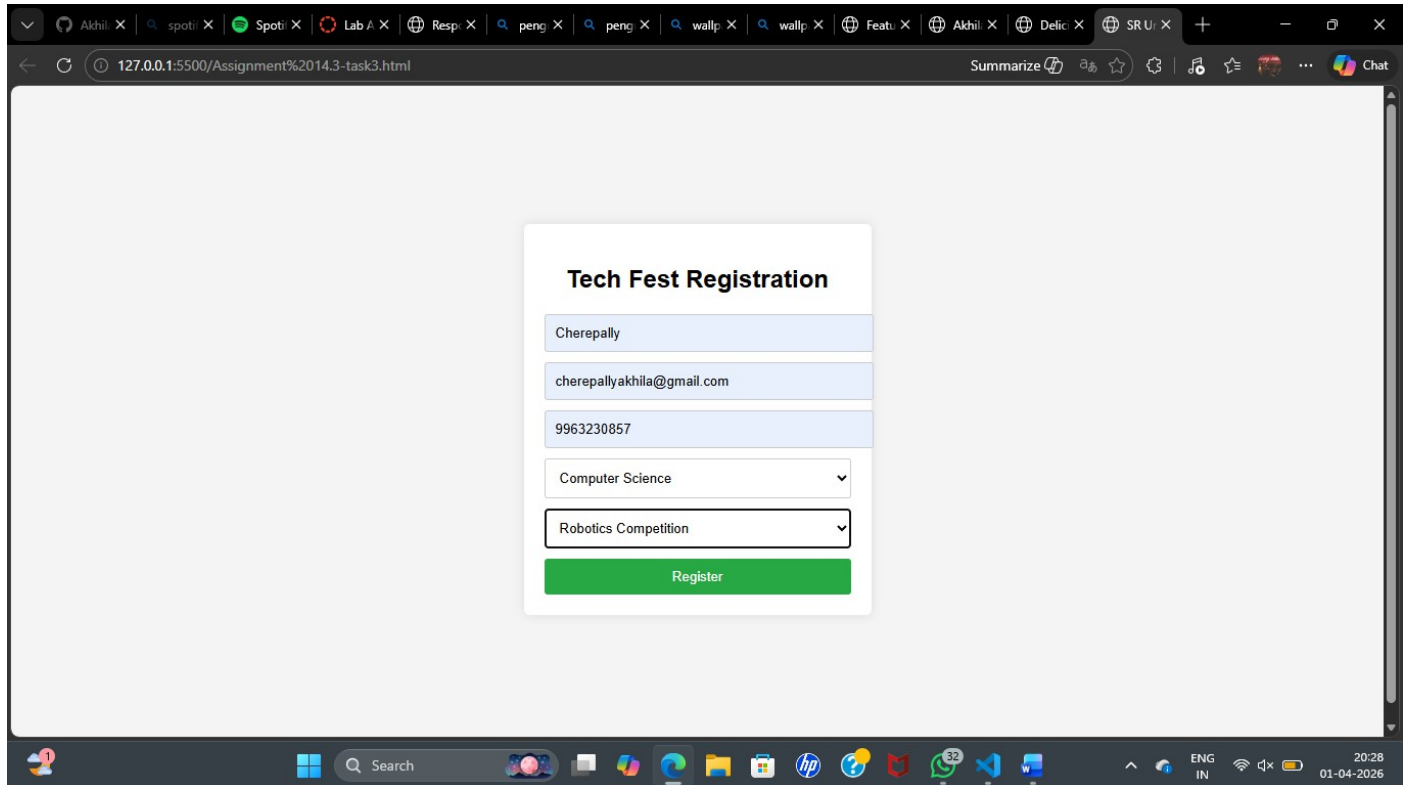
    <select id="event" required>
        <option value="">Select Event</option>
        <option value="coding">Coding Challenge</option>
        <option value="robotics">Robotics Competition</option>
        <option value="hackathon">Hackathon</option>
        <!-- Add more events as needed -->
    </select>
    <div class="error-message" id="eventError"></div>

    <button type="submit">Register</button>
</form>

```

```
<script src="validation.js"></script>
</body>
</html>
```

OUTPUT:



Task 4: AI-Assisted E-Commerce Product Page

Scenario

A startup is developing a basic e-commerce product page to showcase products and manage simple cart functionality.

Tasks

1. Use Copilot to generate a grid-based product catalog using HTML and CSS.
2. Implement an “Add to Cart” feature using JavaScript.
3. Modify AI-generated code to include:
 - o Cart item counter at the top-right corner
 - o Visual feedback when a product is added

Expected Outcome

- Functional product catalog
- Working cart counter
- Interactive user experience

PROMPT: <!--Create a basic e-commerce product page with a grid-based product catalog using HTML and CSS, implement an “Add to Cart” feature with JavaScript, include a cart item counter at the top-right corner, and provide visual feedback when a product is added; ensure the catalog is functional, the cart counter works correctly, and the overall user experience is interactive and responsive.

-->

CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Simple E-commerce Product Page</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #f4f4f4;
      margin: 0;
      padding: 20px;
    }
    .header {
      display: flex;
      justify-content: space-between;
      align-items: center;
      background-color: #333;
      color: #fff;
      padding: 10px 20px;
    }
    .cart-counter {
      background-color: red;
      border-radius: 50%;
      padding: 5px 10px;
      color: #fff;
    }
    .product-grid {
      display: grid;
      grid-template-columns: repeat(auto-fill, minmax(200px, 1fr));
      gap: 20px;
      margin-top: 20px;
    }
    .product-card {
      background-color: #fff;
      padding: 15px;
      border-radius: 5px;
      box-shadow: 0 0 10px rgba(0,0,0,0.1);
      text-align: center;
    }
    .product-card img {
```

```

    max-width: 100%;
    height: auto;
}
.add-to-cart {
    margin-top: 10px;
    padding: 10px 15px;
    background-color: #28a745;
    color: #fff;
    border: none;
    border-radius: 3px;
    cursor: pointer;
}
.add-to-cart:hover {
    background-color: #218838;
}
</style>
</head>
<body>
<div class="header">
    <h1>My E-commerce Store</h1>
    <div class="cart-counter" id="cartCounter">0</div>
</div>

<div class="product-grid">
    <div class="product-card">
        
        <h3>Product 1</h3>
        <p>780.00</p>
        <button class="add-to-cart" onclick="addToCart()">Add to Cart</button>
    </div>
    <div class="product-card">
        
        <h3>Product 2</h3>
        <p>899.00</p>
        <button class="add-to-cart" onclick="addToCart()">Add to Cart</button>
    </div>
    <!-- Add more products as needed i want to add 3 more pics-->
    <div class="product-card">
        
        <h3>Product 3</h3>

```

```

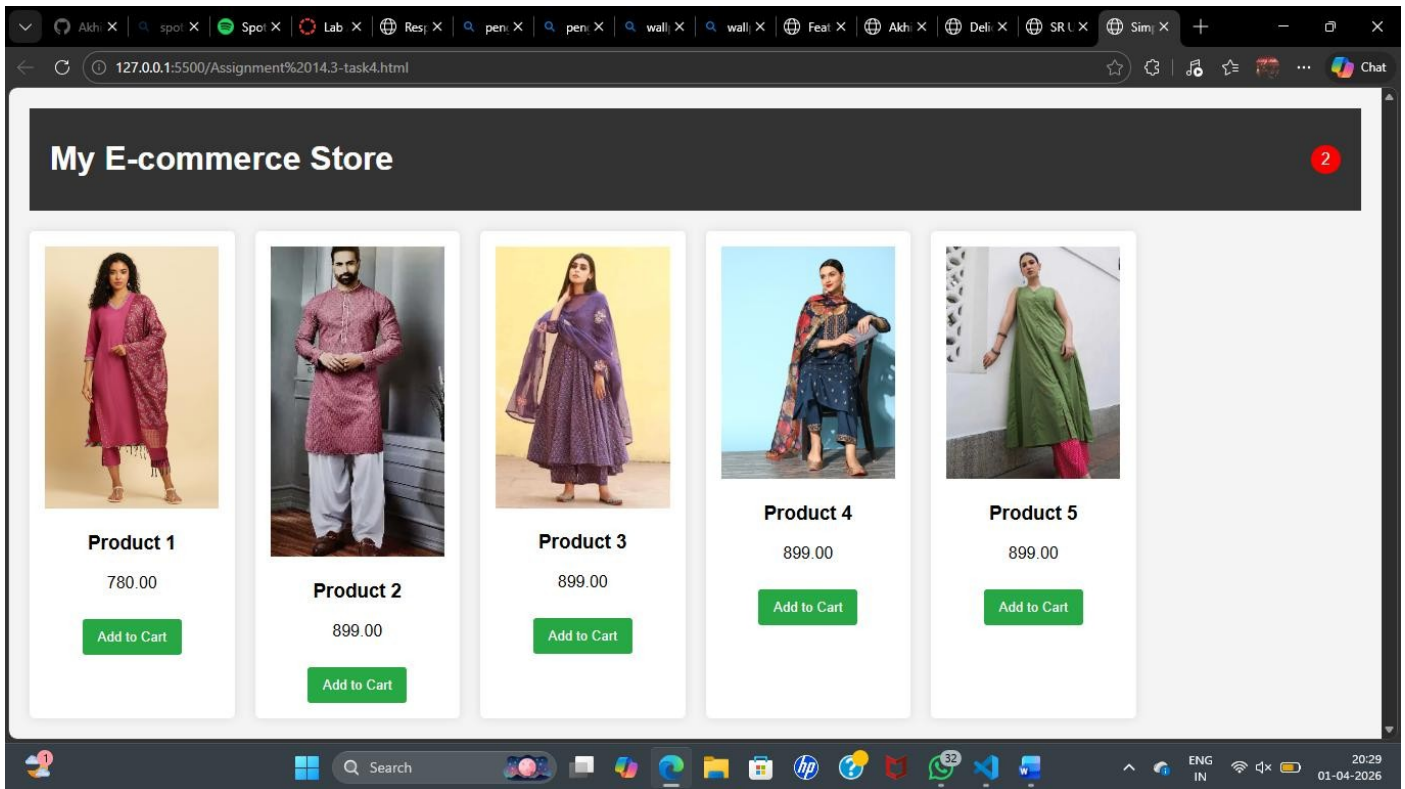
    <p>899.00</p>
    <button class="add-to-cart" onclick="addToCart()">Add to Cart</button>
</div>
<div class="product-card">
    
    <h3>Product 4</h3>
    <p>899.00</p>
    <button class="add-to-cart" onclick="addToCart()">Add to Cart</button>
</div>
<div class="product-card">
    
    <h3>Product 5</h3>
    <p>899.00</p>
    <button class="add-to-cart" onclick="addToCart()">Add to Cart</button>
</div>
</div>

<script>
    let cartCount = 0;

    function addToCart() {
        cartCount++;
        document.getElementById('cartCounter').innerText = cartCount;
        alert('Product added to cart!');
    }
</script>
</body>
</html>

```

OUTPUT:



Task 5: Create a Responsive Web Page Layout

Instructions:

- Design a basic web page layout with a header, main content area, and footer using HTML and CSS.
- Use AI to assist in generating responsive CSS for different screen sizes.
- Ensure the layout is clean and visually organized.

Expected Output:

- A responsive web page with:
- Header with navigation links
- Main section with placeholder text/images
- Footer with copyright or contact info
- Layout adapts correctly to desktop, tablet, and mobile screen sizes.

Note: Report should be submitted as a word document for all tasks in a single document with prompts, comments C code explanation, and output and if required, screenshots.

PROMPT:

```
<!--Design a basic web page layout with a header containing navigation links, a main content area with placeholder text and images, and a footer with copyright or contact info; use HTML and CSS with responsive styling for desktop, tablet, and mobile screen sizes; ensure the layout is clean, visually organized, and adapts correctly across devices with smooth readability and accessibility. English -->
```

CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
```

```
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Responsive Web Page Layout</title>
<style>
  body {
    font-family: Arial, sans-serif;
    margin: 0;
    padding: 0;
    background-color: #f4f4f4;
  }
  header {
    background-color: #333;
    color: #fff;
    padding: 20px;
    text-align: center;
  }
  nav a {
    color: #fff;
    margin: 0 15px;
    text-decoration: none;
  }
  main {
    padding: 20px;
  }
  .placeholder {
    background-color: #ccc;
    height: 200px;
    margin-bottom: 20px;
  }
  footer {
    background-color: #333;
    color: #fff;
    text-align: center;
    padding: 10px;
    position: fixed;
    width: 100%;
    bottom: 0;
  }
  @media (max-width: 768px) {
    nav a {
      display: block;
      margin: 10px 0;
    }
  }
</style>
```

```

</head>
<body>
  <header>
    <h1>ANUV JAIN</h1>
    <nav>
      <a href="#">Home</a>
      <a href="#">About</a>
      <a href="#">Songs</a>
      <a href="#">Contact</a>
    </nav>
  </header>
  <main>
    <div class="image-placeholder">
      
    </div>
  </main>
  <footer>
    Ccopy; 2024 My Responsive Web Page. All rights reserved.
  </footer>

</html>

```

OUTPUT:

